

Unit IV

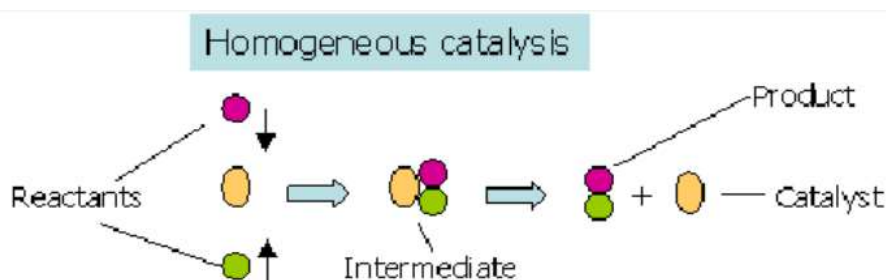
4.3. Homogeneous Catalysis

- Homogeneous catalysis are catalytic reactions in which the catalyst is in the same phase as the reactants. It is generally a catalysis in a solution by a soluble catalyst.
- Proton is the main universal homogeneous catalyst. This is because water is the most common solvent.

4.3.1. Features & Properties

- The catalyst is distributed evenly with the reactants
- The reaction proceeds with the decrease in activation energy
- The reaction continues with the formation of intermediate molecules
- Multiple steps are usually involved in the catalytic process
- They are consumed during a chemical reaction
- They increase the energy of the reactants.
- They offer high degree of interaction between catalyst and reactant molecule.
- They increase the rate of reaction without increase of temperature.
- They shift the equilibrium to favour the formation of products.

4.3.2. Mechanism



Step I: Catalyst-substrate interaction

- The catalytic reaction begins when the catalyst interacts with the reactants to form an intermediate complex.
- The binding usually occurs through coordination bonds, where catalyst is used as a coordination center.

Step II: Reaction intermediates formation

- The crucial step in the catalytic process is the transformation of catalyst-reactant complex into reactive intermediates.
- The reactivity of these intermediates is different than original reactant molecules, which allows the occurrence of desired chemical transformation.

Step III: Catalytic cycle

- This step includes different steps that facilitates the conversion of reactants into desired products; such as substrate binding, activation, reaction process, and release of product.

Step IV: Reaction facilitation

- The catalysts accelerate the chemical reaction by providing an alternate pathway that helps to stabilize the transition states by lowering the energy of activation.

Step V: Catalyst regeneration

- In the final step of catalytic process, after facilitating the reaction, the catalyst is regenerated.

Step VI: Multiple turnovers

- The catalyst can efficiently facilitates the conversion of multiple sets of reactants into products, therefore can repeatedly catalyzed same reaction without being consumed.

4.3.3. Types of homogeneous catalysis:

Type- I

In this type of catalysis, Reactants as well as catalyst are in gaseous state.

- e.g. Oxides of Nitrogen – NO, NO₂ acts as a catalyst for the oxidation of SO₂ to SO₃
- In this process SO₂, O₂ and the catalyst NO, NO₂ all are in the gaseous state.

Type- II

In this type of catalysis, Reactants as well as catalyst are in liquid state.

- e.g. Sulphuric acid is used as a catalyst for the formation of diethyl ether from ethyl alcohol.

Type- III

In this type of catalysis, Reactants as well as catalyst are in solution state.

- e.g. Redox reaction, Iodide ions are oxidized by persulphate ions and Fe(II) or Fe(III) ions in solution acts as catalyst.

Type- IV

In this type of catalysis, acids or bases act as catalyst. According to Bronsted, an acid is a molecule that can give up a proton and base is a molecule that takes up a proton.

Examples

- Acid-Base Catalysis
- Transition Metal Complexes
- Organocatalysis
- Photocatalysis
- Enzymatic Catalysis
- Metathesis Catalysis

4.3.4. Advantages of Homogeneous Catalysis

- Since the reactants and catalyst are in the same phase, the reactions see a higher rate of catalytic activity with more interaction of the catalyst by the reactants.
- These reactions have a higher selectivity in terms of which reactions they catalyze.
- These reactions use milder conditions of reaction as compared to usual.
- Homogeneous catalysts are precisely and easily characterized via spectroscopic methods rather than heterogeneous catalysts in which the reaction takes place with reactants adsorbed on the catalyst surface.
- For exothermic reactions, the homogeneous catalyst passes the heat produced in the reaction into the solvent i.e., dump heat into the solvent.
- Homogeneous catalysts are easier to characterize precisely, so their reaction mechanisms are amenable to rational manipulation.

4.3.5. Disadvantages of Homogeneous Catalysis

- The catalyst, being in the same phase as the reactants, is difficult to isolate from the product at the end of the reaction. In some cases, in which the catalyst is of high activity, removing the catalyst from products might not be possible.
- The separation of homogeneous catalysts from products can be challenging. In some cases, involving high activity catalysts, the catalyst is not removed from the product. In other cases, organic products are sufficiently volatile that they can be separated by distillation

- There is a limitation associated with homogeneous catalysis in terms of thermal stability. Many homogeneous catalysts are organometallic compounds with a higher chance of getting degraded at temperatures above 100 degrees Celsius. Some pincer-based catalysts, however, operate near 200 °C.
- Large amount of water is required for the reaction
- Basic catalysts are affected by the content
- Bulk phase mass transfer limitations are present

4.3.6. Examples of Homogenous catalysis

4.3.6.1. Acid catalysis

In acid catalysis,

- An acid catalyst donates proton(s) (H^+) to the reactant molecules to facilitate the catalytic reaction.
- An electrophilic specie is formed.
- A stable carbocation intermediate is formed.

Examples

- Hydrolysis of Ester or Amines
- Formation of Sulphur dioxide
- Formation of Ethyl acetate

a. Hydrolysis of ester

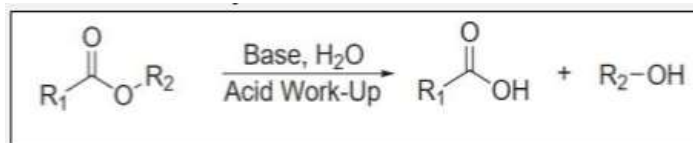
- Acid catalysed hydrolysis reactions involve the cleavage of chemical bonds by the addition of water molecule.
- In the hydrolysis of an ester, the (-OR) group of the ester is replaced by the hydroxyl (-OH) group from water, which results in the formation of a carboxylic acid and an alcohol.



b. Saponification

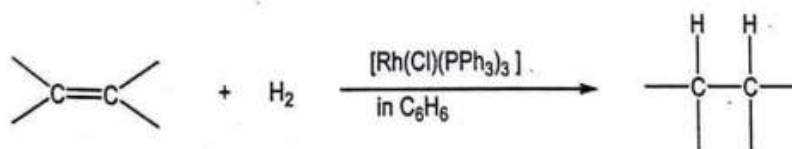
- Base catalyzed hydrolysis reactions involves the cleavage of chemical bonds by the addition of water molecule.

- In the saponification process, the (-OR) group of the ester is replaced by the hydroxyl (-OH) group from water, which results in the formation of a carboxylate ion and an alcohol.



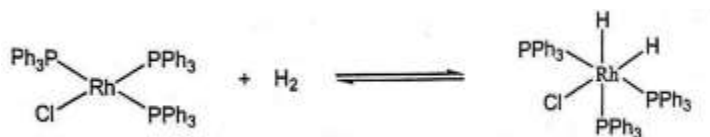
4.3.6.2. Wilkinson's Catalyst

- Wilkinson's catalyst is the familiar name for Chlorotris(tri-phenylphosphine) rhodium (I) - $\text{RhCl}(\text{PPh}_3)_3$
- This reaction has given name after the chemist and Nobel Laureate Sir Geoffrey Wilkinson who popularised its use.
- The compound $\text{RhCl}(\text{PPh}_3)_3$ is a square planar, 16 -electron complex. It is obtained by reaction of rhodium (III) chloride with excess of tri-phenylphosphine.
- Hydrogenation of alkene is a reaction of major industrial importance being used in petrochemical and in the pharmaceutical industries where the preparation of drugs often involves the hydrogenation of specific double bonds.
- The effective homogenous catalysts for this process is $[\text{Rh}(\text{Cl})(\text{PPh}_3)_3]$ chlorotris (triphenylphosphine)rhodium(I).
- This catalyst is known as Wilkinson's catalyst. The effective catalyst for hydrogenation of alkene is $[\text{Rh}(\text{Cl})(\text{PPh}_3)_3]$. The net reaction is

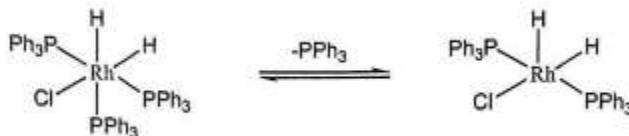


Mechanism of hydrogenation of alkene using Wilkinson's catalyst:

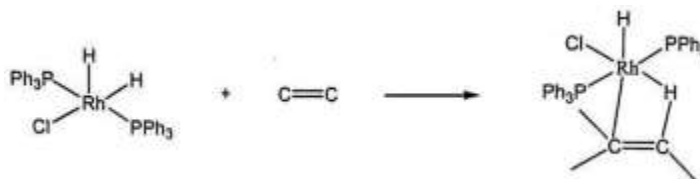
- Step 1: The first step involves the oxidative addition of H_2 to the 16 electron complex $[\text{Rh}(\text{Cl})(\text{PPh}_3)_3]$ to form the 18 electron dihydrido complex.



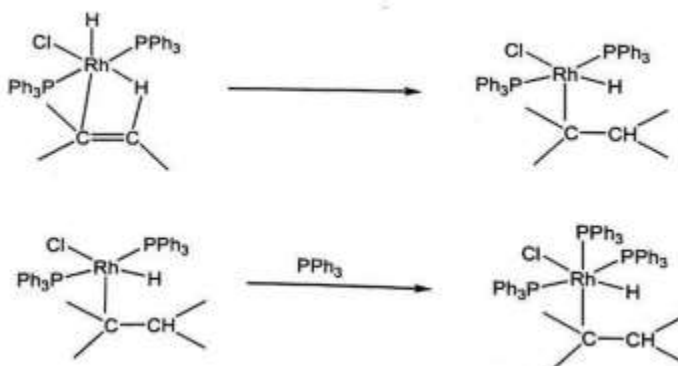
- Step 2: In the second step, dissociation of phosphine ligand results in the formation of the co-ordinately unsaturated complex.



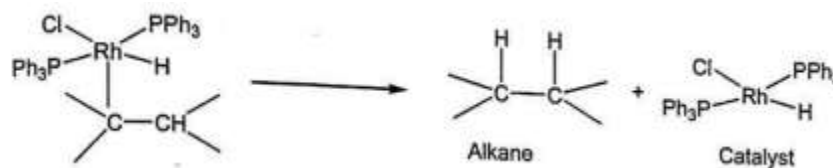
- Step 3: In this step, the coordination of alkene to the potential vacant site, on the 5 coordinate dihydrido complex to six coordinate species.



- Step 4: involves the insertion of olefins into metal hydrogen bond to give metal alkyl derivatives.



- Step 5: is the reductive elimination step, the metal is reduced from +III to +I state and coordination number of the metal is decreased from +6 to +4. In this step the product alkane is formed. The role of the catalyst, namely the transfer of hydrogen fragments to the alkene with the formation of alkane.





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SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA

Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

The catalyst used in this process is being soluble, it can't be removed easily at the end of the reaction by simple filtration. Due to this imperfect separation of alkane and rhodium complex at the end of the hydrogenation, rhodium is lost.